1) a)

$\alpha + \beta = -6$ and $\alpha\beta = -7$

1) b)

$\alpha + \beta = 8$ and $\alpha\beta = 12$

1) c)

$\alpha + \beta = -\frac{3}{2}$ and $\alpha\beta = \frac{9}{2}$

1) d)

$\alpha + \beta = \frac{7}{3}$ and $\alpha\beta = -\frac{8}{3}$

1) e)

$x + 9 = 2x^2 - 10x \Rightarrow 2x^2 - 11x - 9 = 0$

$x^2 - \frac{11}{2}x - \frac{9}{2} = 0$

$\alpha + \beta = \frac{11}{2}$ and $\alpha\beta = -\frac{9}{2}$

2) a)

$ax^2 + bx = 0 \Rightarrow x^2 + \frac{b}{a}x = 0 \Rightarrow x\left(x + \frac{b}{a}\right) = 0$

So $\alpha = 0$ and $\beta = -\frac{b}{a}$

2) b)

$ax^2 + c = 0 \Rightarrow x^2 + \frac{c}{a} = 0 \Rightarrow x^2 = -\frac{c}{a}$

So $\alpha = \sqrt{-\frac{c}{a}}$ and $\beta = -\sqrt{-\frac{c}{a}}$

2) c)

$x^2 + bx + c = 0 \Rightarrow x = \frac{-b \pm \sqrt{b^2 - 4c}}{2}$

So $\alpha = -\frac{b}{2} + \frac{\sqrt{b^2 - 4c}}{2}$ and $\beta = -\frac{b}{2} - \frac{\sqrt{b^2 - 4c}}{2}$

3) a)

$\frac{1}{\alpha}+\frac{1}{\beta}=\frac{\beta}{\alpha\beta}+\frac{\alpha}{\alpha\beta}=\frac{\alpha+\beta}{\alpha\beta}$ as req.

3) b)

Since $(\alpha+\beta)^2=\alpha^2+2\alpha\beta+\beta^2$

$\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta$ as req.

3) c)

$\frac{\alpha}{\beta}+\frac{\beta}{\alpha}=\frac{\alpha^2}{\alpha\beta}+\frac{\beta^2}{\alpha\beta}=\frac{\alpha^2+\beta^2}{\alpha\beta}$

Since $(\alpha+\beta)^2=\alpha^2+2\alpha\beta+\beta^2$

$\frac{\alpha}{\beta}+\frac{\beta}{\alpha}=\frac{(\alpha+\beta)^2-2\alpha\beta}{\alpha\beta}$ as req.

3) d)

$\frac{\alpha^2}{\beta}+\frac{\beta^2}{\alpha}=\frac{\alpha^3}{\alpha\beta}+\frac{\beta^3}{\alpha\beta}=\frac{\alpha^3+\beta^3}{\alpha\beta}$

Since $(\alpha+\beta)^3=\alpha^3+3\alpha\beta^2+3\alpha^2\beta+\beta^3$

$\frac{\alpha^3+\beta^3}{\alpha\beta}=\frac{(\alpha+\beta)^3-3\alpha\beta^2-3\alpha^2\beta}{\alpha\beta}=\frac{(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)}{\alpha\beta}$

$\therefore \frac{\alpha^2}{\beta}+\frac{\beta^2}{\alpha}=\frac{(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)}{\alpha\beta}$ as req.

4) a)

$2x^2+5x-9=0 \Rightarrow x^2+\frac{5}{2}x-\frac{9}{2}=0$

$\alpha+\beta=-\frac{5}{2}$

4) b)

$2x^2+5x-9=0 \Rightarrow x^2+\frac{5}{2}x-\frac{9}{2}=0$

$\alpha\beta=-\frac{9}{2}$

4) c)

$2x^2 + 5x - 9 = 0 \Rightarrow x^2 + \frac{5}{2}x - \frac{9}{2} = 0$

$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\beta}{\alpha\beta} + \frac{\alpha}{\alpha\beta} = \frac{\alpha+\beta}{\alpha\beta} = \frac{-\frac{5}{2}}{-\frac{9}{2}} = \frac{5}{9}$

4) d)

$2x^2 + 5x - 9 = 0 \Rightarrow x^2 + \frac{5}{2}x - \frac{9}{2} = 0$

$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = \left(-\frac{5}{2}\right)^2 - 2\left(-\frac{9}{2}\right) = \frac{61}{4}$

4) e)

$2x^2 + 5x - 9 = 0 \Rightarrow x^2 + \frac{5}{2}x - \frac{9}{2} = 0$

$\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2}{\alpha\beta} + \frac{\beta^2}{\alpha\beta} = \frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{(\alpha+\beta)^2-2\alpha\beta}{\alpha\beta} = \frac{\left(-\frac{5}{2}\right)^2-2\left(-\frac{9}{2}\right)}{-\frac{9}{2}} = -\frac{61}{18}$

4) f)

$2x^2 + 5x - 9 = 0 \Rightarrow x^2 + \frac{5}{2}x - \frac{9}{2} = 0$

$\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha} = \frac{\alpha^3}{\alpha\beta} + \frac{\beta^3}{\alpha\beta} = \frac{\alpha^3+\beta^3}{\alpha\beta} = \frac{(\alpha+\beta)^3-3\alpha\beta^2-3\alpha^2\beta}{\alpha\beta} = \frac{(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)}{\alpha\beta} = \frac{\left(-\frac{5}{2}\right)^3-3\left(-\frac{9}{2}\right)\left(-\frac{5}{2}\right)}{-\frac{9}{2}} = \frac{395}{36}$

5) a)

$7z^2 - 2z + 10 = 0 \Rightarrow z^2 - \frac{2}{7}z + \frac{10}{7} = 0$

$\alpha + \beta = \frac{2}{7}$

5) b)

$7z^2 - 2z + 10 = 0 \Rightarrow z^2 - \frac{2}{7}z + \frac{10}{7} = 0$

$\alpha\beta = \frac{10}{7}$

5) c)

$7z^2 - 2z + 10 = 0 \Rightarrow z^2 - \frac{2}{7}z + \frac{10}{7} = 0$

$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\beta}{\alpha\beta} + \frac{\alpha}{\alpha\beta} = \frac{\alpha+\beta}{\alpha\beta} = \frac{\frac{2}{7}}{\frac{10}{7}} = \frac{1}{5}$

5) d)

$7z^2 - 2z + 10 = 0 \Rightarrow z^2 - \frac{2}{7}z + \frac{10}{7} = 0$

$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = \left(\frac{2}{7}\right)^2 - 2\left(\frac{10}{7}\right) = -\frac{136}{49}$

5) e)

$7z^2 - 2z + 10 = 0 \Rightarrow z^2 - \frac{2}{7}z + \frac{10}{7} = 0$

$\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2}{\alpha\beta} + \frac{\beta^2}{\alpha\beta} = \frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{(\alpha+\beta)^2-2\alpha\beta}{\alpha\beta} = \frac{\left(\frac{2}{7}\right)^2-2\left(\frac{10}{7}\right)}{\frac{10}{7}} = -\frac{68}{35}$

5) f)

$7z^2 - 2z + 10 = 0 \Rightarrow z^2 - \frac{2}{7}z + \frac{10}{7} = 0$

$\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha} = \frac{\alpha^3}{\alpha\beta} + \frac{\beta^3}{\alpha\beta} = \frac{\alpha^3+\beta^3}{\alpha\beta} = \frac{(\alpha+\beta)^3-3\alpha\beta^2-3\alpha^2\beta}{\alpha\beta} = \frac{(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)}{\alpha\beta} = \frac{\left(\frac{2}{7}\right)^3-3\left(\frac{10}{7}\right)\left(\frac{2}{7}\right)}{\frac{10}{7}} = -\frac{206}{245}$

6) a)

$z^2 + 3z + 10 = 0$

$\alpha + \beta = -3$ and $\alpha\beta = 10$

A quadratic with roots α^2 and β^2 requires $\alpha^2 + \beta^2$ and $\alpha^2\beta^2$

$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = (-3)^2 - 2(10) = -11$

$\alpha^2\beta^2 = (\alpha\beta)^2 = 10^2 = 100$

So the equation is

$z^2 + 11z + 100 = 0$

6) b)

$z^2 + 3z + 10 = 0$

$\alpha + \beta = -3$ and $\alpha\beta = 10$

A quadratic with roots $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ requires $\frac{1}{\alpha} + \frac{1}{\beta}$ and $\frac{1}{\alpha\beta}$

$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\beta}{\alpha\beta} + \frac{\alpha}{\alpha\beta} = \frac{\alpha+\beta}{\alpha\beta} = \frac{-3}{10} = -\frac{3}{10}$

$\frac{1}{\alpha\beta} = \frac{1}{10}$

So the equation is

$z^2 + \frac{3}{10}z + \frac{1}{10} = 0 \Rightarrow 10z^2 + 3z + 1 = 0$

7)

$\alpha + \beta = 10$ and $\alpha^2 + \beta^2 = 94$

$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta \Rightarrow 94 = 10^2 - 2\alpha\beta \Rightarrow \alpha\beta = 3$

$\therefore x^2 - 10x + 3 = 0$

So $p = 10$ and $q = 3$

8) a)

$2x^2 + kx - 3 = 0 \Rightarrow x^2 + \frac{k}{2}x - \frac{3}{2} = 0$

$\alpha + \beta = -\frac{k}{2}$

8) b)

$2x^2 + kx - 3 = 0 \Rightarrow x^2 + \frac{k}{2}x - \frac{3}{2} = 0$

$\alpha\beta = -\frac{3}{2}$

8) c)

$2x^2 + kx - 3 = 0 \Rightarrow x^2 + \frac{k}{2}x - \frac{3}{2} = 0$

$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\beta}{\alpha\beta} + \frac{\alpha}{\alpha\beta} = \frac{\alpha+\beta}{\alpha\beta} = \frac{-\frac{k}{2}}{-\frac{3}{2}} = \frac{k}{3}$

8) d)

$2x^2 + kx - 3 = 0 \Rightarrow x^2 + \frac{k}{2}x - \frac{3}{2} = 0$

$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = \left(-\frac{k}{2}\right)^2 - 2\left(-\frac{3}{2}\right) = -\frac{1}{4}k^2 + 3$

9)

$3x^2 - 2x + k = 0 \Rightarrow x^2 - \frac{2}{3}x + \frac{k}{3} = 0$

$\alpha + 3\alpha = \frac{2}{3}$ and $\alpha \times 3\alpha = \frac{k}{3}$

$\therefore \alpha = \frac{1}{6}$ and so $\frac{3}{36} = \frac{k}{3} \Rightarrow k = \frac{1}{4}$

10)

$4x^2 - (9+k)x + k = 0 \Rightarrow x^2 - \left(\frac{9+k}{4}\right)x + \frac{k}{4} = 0$

$\alpha + \alpha + 3 = \frac{9+k}{4}$ and $\alpha(\alpha+3) = \frac{k}{4}$

$\alpha + \alpha + 3 = \frac{9+k}{4} \Rightarrow 8\alpha + 12 = 9 + k \Rightarrow k = 8\alpha + 3$

Substitute into $\alpha(\alpha+3) = \frac{k}{4}$ gives $\alpha(\alpha+3) = \frac{8\alpha+3}{4}$

$4\alpha^2 + 12\alpha = 8\alpha + 3$

$\Rightarrow 4\alpha^2 + 4\alpha - 3 = 0$

So $\alpha = -\frac{3}{2}$ or $\alpha = \frac{1}{2}$

So $k = 8\left(-\frac{3}{2}\right) + 3$ or $k = 8\left(\frac{1}{2}\right) + 3$

$\therefore k = -9$ or $k = 7$

11)

Let the common root be α

From $x^2 - 7x - p = 0$, substitute α in: $\alpha^2 - 7\alpha - p = 0$

From $x^2 + 9x + p = 0$, substitute α in: $\alpha^2 + 9\alpha + p = 0$

$\alpha^2 + 9\alpha + p - (\alpha^2 - 7\alpha - p) = 0$

$16\alpha + 2p = 0$

$\therefore p = -8\alpha$

Substituting back in:

$\alpha^2 - 7\alpha + 8\alpha = 0 \Rightarrow \alpha^2 + \alpha = 0 \Rightarrow \alpha(\alpha+1) = 0$ so $\alpha = 0$ or $\alpha = -1$

So $p = 0$ or $p = 8$ but since p is positive, $p = 8$

12)

$2x^2 + kx + 5 = 0 \Rightarrow x^2 + \frac{k}{2}x + \frac{5}{2} = 0$

$\alpha + \beta = -\frac{k}{2}$ or $\alpha\beta = \frac{5}{2}$

$(\alpha - \beta)^2 = \alpha^2 - 2\alpha\beta + \beta^2$

$\alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta$

so that $(\alpha - \beta)^2 = (\alpha+\beta)^2 - 2\alpha\beta \ - 2\alpha\beta = (\alpha+\beta)^2 - 4\alpha\beta$

$(\alpha - \beta)^2 = \left(-\frac{k}{2}\right)^2 - 4 \times \frac{5}{2} = \frac{1}{4}k^2 - 10$

13)

$\alpha + \beta = 7$ and $\alpha\beta = 10$

$(\alpha + \beta)^3 = \alpha^3 + 3\alpha^2\beta + 3\alpha\beta^3 + \beta^3 = \alpha^3 + \beta^3 + 3\alpha\beta(\alpha + \beta)$

$\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) = 7^3 - 3 \times 10 \times 7 = 133$ as req.

14)

$\alpha + \beta = 4$ and $\alpha\beta = 5$

$(\alpha + \beta)^4 = \alpha^4 + 4\alpha^3\beta + 6\alpha^2\beta^2 + 4\alpha\beta^3 + \beta^4$

$= \alpha^4 + \beta^4 + 6(\alpha\beta)^2 + 4\alpha\beta(\alpha^2 + \beta^2)$

$= \alpha^4 + \beta^4 + 6(\alpha\beta)^2 + 4\alpha\beta((\alpha + \beta)^2 - 2\alpha\beta)$

$\alpha^4 + \beta^4 = (\alpha + \beta)^4 - 6(\alpha\beta)^2 - 4\alpha\beta((\alpha + \beta)^2 - 2\alpha\beta)$

$= 4^4 - 6(5)^2 - 4 \times 5(4^2 - 2 \times 5) = -14$ as req.

15)

Let the roots be α and 3α

Sum of roots: $4\alpha = -\frac{b}{a}$

Product of roots: $3\alpha^2 = \frac{c}{a}$

So that $3\left(-\frac{b}{4a}\right)^2 = \frac{c}{a} \Rightarrow \frac{3b^2}{16a^2} = \frac{c}{a} \Rightarrow 3b^2 = 16ac$ as req.

16)

Let the roots be α and $\alpha + 4$

Sum of roots: $2\alpha + 4 = -\frac{b}{a} \Rightarrow \alpha = -\frac{b}{2a} - 2$

Product of roots: $\alpha(\alpha + 4) = \frac{c}{a}$

So that $\left(-\frac{b}{2a} - 2\right)\left(-\frac{b}{2a} - 2 + 4\right) = \frac{c}{a}$

$\left(-\frac{b}{2a} - 2\right)\left(-\frac{b}{2a} + 2\right) = \frac{c}{a}$

$\frac{b^2}{4a^2} - 4 = \frac{c}{a}$

$\Rightarrow b^2 - 16a^2 = 4ac$ as req.

17)

$\alpha + \beta = 6$ and $\alpha\beta = 8$

Sum of roots: $\alpha^3 + \beta + \alpha + \beta^3$

Product of roots: $(\alpha^3 + \beta)(\beta^3 + \alpha)$

$\alpha^3 + \beta + \alpha + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) + \alpha + \beta = 6^3 - 3 \times 8(6) + 6 = 78$

$(\alpha^3 + \beta)(\beta^3 + \alpha) = \alpha^3\beta^3 + \beta^4 + \alpha^4 + \alpha\beta$

$= (\alpha\beta)^3 + (\alpha + \beta)^4 - 6(\alpha\beta)^2 - 4\alpha\beta((\alpha + \beta)^2 - 2\alpha\beta) + \alpha\beta$

$= 8^3 + 6^4 - 6 \times 8^2 - 4 \times 8(6^2 - 2 \times 8) + 8 = 792$

$\therefore x^2 - 78x + 792 = 0$

18)

For f the sum of roots is $\frac{24}{p}$

For g the product of roots is $\frac{3}{q}$

$\frac{24}{p} = \frac{3}{q}$

$\Rightarrow 8q = p$

19)

Sum of roots: $2\beta + \beta = 3\beta = -p$

Product of roots: $2\alpha\beta = 2\beta^2 = q$

So $\beta = -\frac{p}{3}$ and so $2\left(-\frac{p}{3}\right)^2 = q$

$\therefore 2p^2 = 9q$ as req.

20)

Sum of roots: $\alpha + 5 + \mathrm{i} = k$

Product of roots: $\alpha(5 + \mathrm{i}) = 25 + 18\mathrm{i}$

$\alpha = \frac{25+18\mathrm{i}}{5+\mathrm{i}} \times \frac{5-\mathrm{i}}{5-\mathrm{i}} = \frac{143+65\mathrm{i}}{26} = \frac{11}{2} + \frac{5}{2}\mathrm{i}$

So $\frac{11}{2} + \frac{5}{2}\mathrm{i} + 5 + \mathrm{i} = k$

$\therefore k = \frac{21}{2} + \frac{7}{2}\mathrm{i}$

21) a)

Sum of roots: $\alpha + \beta = -\frac{b}{a}$

Product of roots: $\alpha\beta = \frac{c}{a}$

New sum of roots: $\alpha + \beta + \alpha\beta = -\frac{b}{a} + \frac{c}{a}$

New product of roots: $\alpha\beta(\alpha + \beta) = \frac{c}{a} \times -\frac{b}{a}$

$x^2 - \left(-\frac{b}{a} + \frac{c}{a}\right)x - \frac{cb}{a^2} = 0$

$ax^2 + a(b - c)x - bc = 0$

21) b)

Two real distinct roots requires a positive discriminant. So:

$\big(a(b - c)\big)^2 - 4a(-bc)$

$= a^2(b - c)^2 + 4a^2bc$

Since a, b and c are all positive integers

$a^2 > 0$, $(b - c)^2 \geq 0$ and $bc > 0$

And so $a^2(b - c)^2 + 4a^2bc > 0$

Therefore the discriminant is positive and so there are two distinct real roots.

22) a)

$\alpha\beta = k$

22) b)

$\alpha + \beta = 6$

22) c)

Sum of roots: $\alpha - 2 + \beta + 2 = \alpha + \beta = 6$

Product of roots: $(\alpha - 2)(\beta + 2) = \alpha\beta + 2\alpha - 2\beta - 4$

$\alpha\beta + 2\alpha - 2\beta - 4 = k + 2(\alpha - \beta) - 4$

Use the solutions of $x^2 - 6x + k = 0$ to find the difference between α and β:

$\frac{6 \pm \sqrt{36 - 4k}}{2}$ so $\alpha - \beta = 3 + \frac{\sqrt{36 - 4k}}{2} - \left(3 - \frac{\sqrt{36 - 4k}}{2}\right) = \sqrt{36 - 4k}$

$\therefore\ k + 2(\alpha - \beta) - 4 = k + 2\sqrt{36 - 4k} - 4$

So the final quadratic is:

$x^2 - 6x + \big(k - 4 + 2\sqrt{36 - 4k}\ \big) = 0$ as req.

23) a)

$\alpha + \beta + \gamma = -3, \alpha\beta + \alpha\gamma + \beta\gamma = -5, \alpha\beta\gamma = 1$

23) b)

$\alpha + \beta + \gamma = 9, \alpha\beta + \alpha\gamma + \beta\gamma = 5, \alpha\beta\gamma = -4$

23) c)

$2z^3 + 5z^2 + 11z - 2 = 0 \Rightarrow z^3 + \frac{5}{2}z^2 + \frac{11}{2}z - 1 = 0$

$\alpha + \beta + \gamma = -\frac{5}{2}, \alpha\beta + \alpha\gamma + \beta\gamma = \frac{11}{2}, \alpha\beta\gamma = 1$

23) d)

$-5z^3 + z^2 - 12 = 0 \Rightarrow z^3 - \frac{1}{5}z^2 + \frac{12}{5} = 0$

$\alpha + \beta + \gamma = \frac{1}{5}, \alpha\beta + \alpha\gamma + \beta\gamma = 0, \alpha\beta\gamma = -\frac{12}{5}$

23) e)

$x + 1 = 4x(x - 1)(x + 2)$

$x + 1 = 4x(x^2 + x - 2)$

$x + 1 = 4x^3 + 4x^2 - 8x$

$4x^3 + 4x^2 - 9x - 1 = 0$

$\Rightarrow x^3 + x^2 - \frac{9}{4}x - \frac{1}{4} = 0$

$\alpha + \beta + \gamma = -1, \alpha\beta + \alpha\gamma + \beta\gamma = -\frac{9}{4}, \alpha\beta\gamma = \frac{1}{4}$

24)

$z^3 - 3z^2 + 8z + 4 = 0$

But the z^3 has coefficient of 2 so $f(z) = 2z^3 - 6z^2 + 16z + 8$

25) a)

Sum of roots: $2 - 1 + \alpha = -6$

So $\alpha = -7$

25) b)

Sum of pairs of products of roots: $-2 + 2\alpha - \alpha = -p$

$-2 - 14 + 7 = -p \therefore p = 9$

Product of roots: $-2\alpha = q \therefore q = 14$

26) a)

$2z^3 + pz^2 - 4z + q = 0 \Rightarrow z^3 + \frac{p}{2}z^2 - 4z + \frac{q}{2} = 0$

Sum of roots: $\alpha + \beta + \gamma = -\frac{p}{2}$ so $p = 3$

Product of roots: $\alpha\beta\gamma = -10 = -\frac{q}{2}$ so $q = 20$

26) b)

$2z^3 + 3z^2 - 4z + 20 = 0$

$z \in \{-3.146, 0.823 - 1.582\text{i}, 0.823 + 1.582\text{i}\}$ (3 DP)

27) a)

$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma}{\alpha\beta\gamma} + \frac{\alpha\gamma}{\alpha\beta\gamma} + \frac{\alpha\beta}{\alpha\beta\gamma} = \frac{\alpha\beta+\alpha\gamma+\beta\gamma}{\alpha\beta\gamma}$ as req.

27) b)

Since $(\alpha + \beta + \gamma)^2 = (\alpha + \beta + \gamma)(\alpha + \beta + \gamma)$

$= \alpha^2 + \alpha\beta + \alpha\gamma + \alpha\beta + \beta^2 + \beta\gamma + \alpha\gamma + \beta\gamma + \gamma^2$

$= \alpha^2 + \beta^2 + \gamma^2 + 2\alpha\beta + 2\alpha\gamma + 2\beta\gamma$

So that $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$ as req.

27) c)

Since $(\alpha + \beta + \gamma)(\alpha\beta + \alpha\gamma + \beta\gamma)$

$= \alpha^2\beta + \alpha^2\gamma + \alpha\beta\gamma + \alpha\beta^2 + \alpha\beta\gamma + \beta^2\gamma + \alpha\beta\gamma + \alpha\gamma^2 + \beta\gamma^2$

$= \alpha^2\beta + \alpha^2\gamma + \alpha\beta^2 + \beta^2\gamma + \alpha\gamma^2 + \beta\gamma^2 + 3\alpha\beta\gamma$

And

$(\alpha + \beta)(\alpha + \gamma)(\beta + \gamma) = (\alpha + \beta)(\alpha\beta + \alpha\gamma + \beta\gamma + \gamma^2)$

$= \alpha^2\beta + \alpha^2\gamma + \alpha\beta\gamma + \alpha\gamma^2 + \alpha\beta^2 + \alpha\beta\gamma + \beta^2\gamma + \beta\gamma^2$

$= \alpha^2\beta + \alpha^2\gamma + \alpha\gamma^2 + \alpha\beta^2 + \beta^2\gamma + \beta\gamma^2 + 2\alpha\beta\gamma$

$= (\alpha + \beta + \gamma)(\alpha\beta + \alpha\gamma + \beta\gamma) - \alpha\beta\gamma$ as req.

27) d)

Since $(\alpha\beta + \alpha\gamma + \beta\gamma)^2 = (\alpha\beta + \alpha\gamma + \beta\gamma)(\alpha\beta + \alpha\gamma + \beta\gamma)$

$= \alpha^2\beta^2 + \alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha^2\beta\gamma + \alpha^2\gamma^2 + \alpha\beta\gamma^2 + \alpha\beta^2\gamma + \alpha\beta\gamma^2 + \beta^2\gamma^2$

$= \alpha^2\beta^2 + \beta^2\gamma^2 + \alpha^2\gamma^2 + 2\alpha^2\beta\gamma + 2\alpha\beta^2\gamma + 2\alpha\beta\gamma^2$

So that $\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2 = (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2(\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2)$

$= (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)$

28) a)

$\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2 = \alpha\beta\gamma(\alpha + \beta + \gamma)$

28) b)

Since $(\alpha\beta + \alpha\gamma + \beta\gamma)^2 = (\alpha\beta + \alpha\gamma + \beta\gamma)(\alpha\beta + \alpha\gamma + \beta\gamma)$

$= \alpha^2\beta^2 + \alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha^2\beta\gamma + \alpha^2\gamma^2 + \alpha\beta\gamma^2 + \alpha\beta^2\gamma + \alpha\beta\gamma^2 + \beta^2\gamma^2$

$= \alpha^2\beta^2 + \beta^2\gamma^2 + \alpha^2\gamma^2 + 2\alpha^2\beta\gamma + 2\alpha\beta^2\gamma + 2\alpha\beta\gamma^2$

So that $\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2 = (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2(\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2)$

$= (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)$

So

$\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2 = (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)$

29) a)

$3x^3 + 4x^2 - 8x - 2 = 0 \Rightarrow x^3 + \frac{4}{3}x^2 - \frac{8}{3}x - \frac{2}{3} = 0$

$\alpha + \beta + \gamma = -\frac{4}{3}$

29) b)

$3x^3 + 4x^2 - 8x - 2 = 0 \Rightarrow x^3 + \frac{4}{3}x^2 - \frac{8}{3}x - \frac{2}{3} = 0$

$\alpha\beta + \alpha\gamma + \beta\gamma = -\frac{8}{3}$

29) c)

$3x^3 + 4x^2 - 8x - 2 = 0 \Rightarrow x^3 + \frac{4}{3}x^2 - \frac{8}{3}x - \frac{2}{3} = 0$

$\alpha\beta\gamma = \frac{2}{3}$

29) d)

$3x^3 + 4x^2 - 8x - 2 = 0 \Rightarrow x^3 + \frac{4}{3}x^2 - \frac{8}{3}x - \frac{2}{3} = 0$

$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma}{\alpha\beta\gamma} + \frac{\alpha\gamma}{\alpha\beta\gamma} + \frac{\alpha\beta}{\alpha\beta\gamma} = \frac{\alpha\beta+\alpha\gamma+\beta\gamma}{\alpha\beta\gamma}$

$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{-\frac{8}{3}}{\frac{2}{3}} = -4$

29) e)

$3x^3 + 4x^2 - 8x - 2 = 0 \Rightarrow x^3 + \frac{4}{3}x^2 - \frac{8}{3}x - \frac{2}{3} = 0$

$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$

$\alpha^2 + \beta^2 + \gamma^2 = \left(-\frac{4}{3}\right)^2 - 2\left(-\frac{8}{3}\right) = \frac{64}{9}$

29) f)

$3x^3 + 4x^2 - 8x - 2 = 0 \Rightarrow x^3 + \frac{4}{3}x^2 - \frac{8}{3}x - \frac{2}{3} = 0$

$\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2 = \alpha\beta\gamma(\alpha + \beta + \gamma)$

$\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2 = \frac{2}{3}\left(-\frac{4}{3}\right) = -\frac{8}{9}$

29) g)

$3x^3 + 4x^2 - 8x - 2 = 0 \Rightarrow x^3 + \frac{4}{3}x^2 - \frac{8}{3}x - \frac{2}{3} = 0$

$\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2 = (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2(\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2)$

$= (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)$

$\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2 = \left(-\frac{8}{3}\right)^2 - 2 \times \frac{2}{3}\left(-\frac{4}{3}\right) = \frac{80}{9}$

29) h)

$3x^3 + 4x^2 - 8x - 2 = 0 \Rightarrow x^3 + \frac{4}{3}x^2 - \frac{8}{3}x - \frac{2}{3} = 0$

$(\alpha + \beta)(\alpha + \gamma)(\beta + \gamma) = (\alpha + \beta)(\alpha\beta + \alpha\gamma + \beta\gamma + \gamma^2)$

$= \alpha^2\beta + \alpha^2\gamma + \alpha\beta\gamma + \alpha\gamma^2 + \alpha\beta^2 + \alpha\beta\gamma + \beta^2\gamma + \beta\gamma^2$

$= \alpha^2\beta + \alpha^2\gamma + \alpha\gamma^2 + \alpha\beta^2 + \beta^2\gamma + \beta\gamma^2 + 2\alpha\beta\gamma$

$= (\alpha + \beta + \gamma)(\alpha\beta + \alpha\gamma + \beta\gamma) - \alpha\beta\gamma$

$(\alpha + \beta)(\alpha + \gamma)(\beta + \gamma) = \left(-\frac{4}{3}\right)\left(-\frac{8}{3}\right) - \frac{2}{3} = \frac{26}{9}$

29) i)

$3x^3 + 4x^2 - 8x - 2 = 0 \Rightarrow x^3 + \frac{4}{3}x^2 - \frac{8}{3}x - \frac{2}{3} = 0$

$\frac{\alpha^2\beta^2+\alpha^2\gamma^2+\beta^2\gamma^2}{\alpha\beta\gamma} = \frac{(\alpha\beta+\alpha\gamma+\beta\gamma)^2-2\alpha\beta\gamma(\alpha+\beta+\gamma)}{\alpha\beta\gamma}$

$\frac{\alpha^2\beta^2+\alpha^2\gamma^2+\beta^2\gamma^2}{\alpha\beta\gamma} = \frac{\left(-\frac{8}{3}\right)^2-2\times\frac{2}{3}\left(-\frac{4}{3}\right)}{\frac{2}{3}} = \frac{40}{3}$

30) a)

$5z^3 - 2z^2 + z + 1 = 0 \Rightarrow z^3 - \frac{2}{5}z^2 + \frac{1}{5}z + \frac{1}{5} = 0$

$\alpha + \beta + \gamma = \frac{2}{5}$

30) b)

$5z^3 - 2z^2 + z + 1 = 0 \Rightarrow z^3 - \frac{2}{5}z^2 + \frac{1}{5}z + \frac{1}{5} = 0$

$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{1}{5}$

30) c)

$5z^3 - 2z^2 + z + 1 = 0 \Rightarrow z^3 - \frac{2}{5}z^2 + \frac{1}{5}z + \frac{1}{5} = 0$

$\alpha\beta\gamma = -\frac{1}{5}$

30) d)

$5z^3 - 2z^2 + z + 1 = 0 \Rightarrow z^3 - \frac{2}{5}z^2 + \frac{1}{5}z + \frac{1}{5} = 0$

$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma}{\alpha\beta\gamma} + \frac{\alpha\gamma}{\alpha\beta\gamma} + \frac{\alpha\beta}{\alpha\beta\gamma} = \frac{\alpha\beta+\alpha\gamma+\beta\gamma}{\alpha\beta\gamma}$

$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\frac{1}{5}}{-\frac{1}{5}} = -1$

30) e)

$5z^3 - 2z^2 + z + 1 = 0 \Rightarrow z^3 - \frac{2}{5}z^2 + \frac{1}{5}z + \frac{1}{5} = 0$

$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$

$\alpha^2 + \beta^2 + \gamma^2 = \left(\frac{2}{5}\right)^2 - 2\left(\frac{1}{5}\right) = -\frac{6}{25}$

30) f)

$5z^3 - 2z^2 + z + 1 = 0 \Rightarrow z^3 - \frac{2}{5}z^2 + \frac{1}{5}z + \frac{1}{5} = 0$

$\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2 = \alpha\beta\gamma(\alpha + \beta + \gamma)$

$\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2 = -\frac{1}{5}\left(\frac{2}{5}\right) = -\frac{2}{25}$

30) g)

$5z^3 - 2z^2 + z + 1 = 0 \Rightarrow z^3 - \frac{2}{5}z^2 + \frac{1}{5}z + \frac{1}{5} = 0$

$\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2 = (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2(\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2)$

$= (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)$

$\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2 = \left(\frac{1}{5}\right)^2 - 2 \times -\frac{1}{5}\left(\frac{2}{5}\right) = \frac{1}{5}$

30) h)

$5z^3 - 2z^2 + z + 1 = 0 \Rightarrow z^3 - \frac{2}{5}z^2 + \frac{1}{5}z + \frac{1}{5} = 0$

$(\alpha + \beta)(\alpha + \gamma)(\beta + \gamma) = (\alpha + \beta)(\alpha\beta + \alpha\gamma + \beta\gamma + \gamma^2)$

$= \alpha^2\beta + \alpha^2\gamma + \alpha\beta\gamma + \alpha\gamma^2 + \alpha\beta^2 + \alpha\beta\gamma + \beta^2\gamma + \beta\gamma^2$

$= \alpha^2\beta + \alpha^2\gamma + \alpha\gamma^2 + \alpha\beta^2 + \beta^2\gamma + \beta\gamma^2 + 2\alpha\beta\gamma$

$= (\alpha + \beta + \gamma)(\alpha\beta + \alpha\gamma + \beta\gamma) - \alpha\beta\gamma$

$(\alpha + \beta)(\alpha + \gamma)(\beta + \gamma) = \left(\frac{2}{5}\right)\left(\frac{1}{5}\right) + \frac{1}{5} = \frac{7}{25}$

30) i)

$5z^3 - 2z^2 + z + 1 = 0 \Rightarrow z^3 - \frac{2}{5}z^2 + \frac{1}{5}z + \frac{1}{5} = 0$

$\frac{\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2}{\alpha\beta\gamma} = \frac{(\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)}{\alpha\beta\gamma}$

$\frac{\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2}{\alpha\beta\gamma} = \frac{\left(\frac{1}{5}\right)^2 - 2 \times -\frac{1}{5}\left(\frac{2}{5}\right)}{-\frac{1}{5}} = -1$

31)

Since there is no x^2 term $\alpha + \beta + \gamma = 0$ and so $\alpha + \beta = -\gamma$

32)

$\alpha\beta + \alpha\gamma + \beta\gamma = 0$ and $\alpha\beta\gamma = -5p$

Let $\alpha\beta = 12$ so that $12\gamma = -5p$

$\therefore \gamma = -\frac{5p}{12}$ as req.

33)

$6x^3 + kx^2 - 26x + 15 = 0 \Rightarrow x^3 + \frac{k}{6}x^2 - \frac{13}{3}x + \frac{5}{2} = 0$

Sum of roots: $\alpha + \beta + \gamma = \frac{11}{6}$

$-\frac{k}{6} = \frac{11}{6} \therefore k = -11$

34) a)

$10x^3 - 57x^2 + kx - 5 = 0 \Rightarrow x^3 - \frac{57}{10}x^2 + \frac{k}{10}x - \frac{1}{2} = 0$

Product of roots: $\alpha\beta\left(\frac{1}{\alpha}\right) = \frac{1}{2}$

So $\beta = \frac{1}{2}$

Sum of roots: $\alpha + \frac{1}{\alpha} + \beta = \frac{57}{10}$

$\alpha + \frac{1}{\alpha} + \frac{1}{2} = \frac{57}{10}$

$\alpha + \frac{1}{\alpha} = \frac{52}{10}$

$10\alpha^2 - 52\alpha + 10 = 0 \therefore \alpha = 5$ or $\alpha = \frac{1}{5}$

So the solutions are $5, \frac{1}{5}$ and $\frac{1}{2}$

34) b)

Sum of pairs of products of roots: $\alpha\left(\frac{1}{\alpha}\right) + \alpha\beta + \beta\left(\frac{1}{\alpha}\right) = \frac{k}{10}$

$5\left(\frac{1}{5}\right) + 5 \times \frac{1}{2} + \frac{1}{2}\left(\frac{1}{5}\right) = \frac{k}{10}$

$\therefore k = 36$

35) a)

$z^3 - 4z^2 + 3z - 7 = 0$ so $\alpha + \beta + \gamma = 4$, $\alpha\beta + \alpha\gamma + \beta\gamma = 3$ and $\alpha\beta\gamma = 7$

$\frac{2}{\alpha} + \frac{2}{\beta} + \frac{2}{\gamma} = 2\left(\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}\right)$

Since $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma}{\alpha\beta\gamma} + \frac{\alpha\gamma}{\alpha\beta\gamma} + \frac{\alpha\beta}{\alpha\beta\gamma} = \frac{\alpha\beta+\alpha\gamma+\beta\gamma}{\alpha\beta\gamma}$

$\frac{2}{\alpha} + \frac{2}{\beta} + \frac{2}{\gamma} = 2\left(\frac{\alpha\beta+\alpha\gamma+\beta\gamma}{\alpha\beta\gamma}\right) = 2\left(\frac{3}{7}\right) = \frac{6}{7}$

35) b)

$z^3 - 4z^2 + 3z - 7 = 0$ so $\alpha + \beta + \gamma = 4$, $\alpha\beta + \alpha\gamma + \beta\gamma = 3$ and $\alpha\beta\gamma = 7$

$(\alpha + 3)(\beta + 3)(\gamma + 3) = (\alpha + 3)(\beta\gamma + 3\beta + 3\gamma + 9)$

$= \alpha\beta\gamma + 3\alpha\beta + 3\alpha\gamma + 9\alpha + 3\beta\gamma + 9\beta + 9\gamma + 27$

$= \alpha\beta\gamma + 3(\alpha\beta + \alpha\gamma + \beta\gamma) + 9(\alpha + \beta + \gamma) + 27$

So $(\alpha + 3)(\beta + 3)(\gamma + 3) = 7 + 3(3) + 9(4) + 27 = 79$

35) c)

$z^3 - 4z^2 + 3z - 7 = 0$ so $\alpha + \beta + \gamma = 4$, $\alpha\beta + \alpha\gamma + \beta\gamma = 3$ and $\alpha\beta\gamma = 7$

$\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2 = (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2(\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2)$

$= (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)$

So that

$\frac{\alpha^2\beta^2+\alpha^2\gamma^2+\beta^2\gamma^2}{5\alpha\beta\gamma} = \frac{(\alpha\beta+\alpha\gamma+\beta\gamma)^2-2\alpha\beta\gamma(\alpha+\beta+\gamma)}{5\alpha\beta\gamma} = \frac{3^2-2\times7(4)}{5\times7} = -\frac{47}{35}$

35) d)

$z^3 - 4z^2 + 3z - 7 = 0$ so $\alpha + \beta + \gamma = 4$, $\alpha\beta + \alpha\gamma + \beta\gamma = 3$ and $\alpha\beta\gamma = 7$

$(\alpha + \beta)(\alpha + \gamma)(\beta + \gamma) = (\alpha + \beta)(\alpha\beta + \alpha\gamma + \beta\gamma + \gamma^2)$

$= \alpha^2\beta + \alpha^2\gamma + \alpha\beta\gamma + \alpha\gamma^2 + \alpha\beta^2 + \alpha\beta\gamma + \beta^2\gamma + \beta\gamma^2$

$= \alpha^2\beta + \alpha^2\gamma + \alpha\gamma^2 + \alpha\beta^2 + \beta^2\gamma + \beta\gamma^2 + 2\alpha\beta\gamma$

$= (\alpha + \beta + \gamma)(\alpha\beta + \alpha\gamma + \beta\gamma) - \alpha\beta\gamma$

$(2\alpha + 2\beta)(2\alpha + 2\gamma)(2\beta + 2\gamma) = 8(\alpha + \beta)(\alpha + \gamma)(\beta + \gamma)$

$= 8[(\alpha + \beta + \gamma)(\alpha\beta + \alpha\gamma + \beta\gamma) - \alpha\beta\gamma]$

$= 8[4 \times 3 - 7] = 40$

36)

$x^3 + kx^2 - 5x + 3 = 0$

Sum of roots: $\alpha + \beta + \gamma = -k$

Sum of pairs of products of roots: $\alpha\beta + \alpha\gamma + \beta\gamma = 5$

Product of roots: $\alpha\beta\gamma = -3$

$\alpha^2\beta^2 + \alpha^2\beta^2 + \beta^2\gamma^2 = (\alpha\beta + \beta\gamma + \alpha\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)$

$= 5^2 - 2(-k)(-3) = 25 - 6k$ as req.

37) a)

The sum of the squares of the roots is negative, which is impossible if all three roots are real because the square of any real number is non-negative and the sum of three non-negative numbers cannot be negative. So there must be one real root and two complex roots (which are complex conjugates of each other).

37) b)

$z^3 - 7z^2 + kz - 120 = 0$

Sum of roots: $\alpha + \beta + \gamma = 7$

Sum of pairs of products of roots: $\alpha\beta + \alpha\gamma + \beta\gamma = k$

Product of roots: $\alpha\beta\gamma = 120$

$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$

$(\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) = -55$

$7^2 - 2k = -55$

$\therefore k = 52$

38)

$z^3 + kz + 35 - 5\mathrm{i} = 0$

Sum of roots: $\alpha + \beta + \gamma = 0$

Since $\alpha + \beta = 5$ then $\gamma = -5$

Product of roots: $\alpha\beta\gamma = -35 + 5\mathrm{i}$

$\therefore \alpha\beta = \frac{-35+5\mathrm{i}}{-5} = 7 - \mathrm{i}$

Sum of pairs of products of roots: $\alpha\beta + \alpha\gamma + \beta\gamma = k$

$\alpha\beta - 5\alpha - 5\beta = k$

$7 - \mathrm{i} - 5(\alpha + \beta) = k$

$7 - \mathrm{i} - 5 \times 5 = k$

$\therefore k = -18 - \mathrm{i}$

39)

$k^3 + kz^2 + 6 + \mathrm{i} = 0$

Sum of roots: $\alpha + \beta + \gamma = k$

Sum of pairs of products of roots: $\alpha\beta + \alpha\gamma + \beta\gamma = 0$

But $\alpha^2 + \beta^2 + \gamma^2 = 2\mathrm{i}$

$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$

$k^2 - 2(0) = 2\mathrm{i}$

$k^2 = 2\mathrm{i}$

Let $k = a + b\mathrm{i}$

$(a + b\mathrm{i})^2 = 2\mathrm{i}$

$a^2 + 2ab\mathrm{i} - b^2 = 2\mathrm{i}$

$\therefore a^2 - b^2 = 0$ and $2ab = 2 \Rightarrow a = \frac{1}{b}$

$\left(\frac{1}{b}\right)^2 - b^2 = 0 \Rightarrow 1 - b^4 = 0$

$b = 1$ or $b = -1$ and so $a = 1$ or $a = -1$

So $k = -1 - \mathrm{i}$ or $k = 1 + \mathrm{i}$

40) a)

Since $(\alpha+\beta+\gamma)^3=(\alpha+\beta+\gamma)(\alpha^2+\beta^2+\gamma^2+2\alpha\beta+2\alpha\gamma+2\beta\gamma)$

$=\alpha^3+\alpha\beta^2+\alpha\gamma^2+2\alpha^2\beta+2\alpha^2\gamma+2\alpha\beta\gamma+\alpha^2\beta+\beta^3+\beta\gamma^2+2\alpha\beta^2+2\alpha\beta\gamma+$
$2\beta^2\gamma+\alpha^2\gamma+\beta^2\gamma+\gamma^3+2\alpha\beta\gamma+2\alpha\gamma^2+2\beta\gamma^2$

$=\alpha^3+\beta^3+\gamma^3+3\alpha\beta^2+3\alpha\gamma^2+3\alpha^2\beta+3\alpha^2\gamma+6\alpha\beta\gamma+3\beta\gamma^2+3\beta^2\gamma$

$=\alpha^3+\beta^3+\gamma^3+3(\alpha\beta^2+\alpha\gamma^2+\alpha^2\beta+\alpha^2\gamma+\beta\gamma^2+\beta^2\gamma+2\alpha\beta\gamma)$

But given

$\alpha^2\beta+\alpha^2\gamma+\alpha\gamma^2+\alpha\beta^2+\beta^2\gamma+\beta\gamma^2+2\alpha\beta\gamma=(\alpha+\beta+\gamma)(\alpha\beta+\alpha\gamma+\beta\gamma)-\alpha\beta\gamma$

This gives

$(\alpha+\beta+\gamma)^3=\alpha^3+\beta^3+\gamma^3+3\big((\alpha+\beta+\gamma)(\alpha\beta+\alpha\gamma+\beta\gamma)-\alpha\beta\gamma\big)$

So $\alpha^3+\beta^3+\gamma^3=(\alpha+\beta+\gamma)^3-3\big((\alpha+\beta+\gamma)(\alpha\beta+\alpha\gamma+\beta\gamma)-\alpha\beta\gamma\big)$

40) b)

$2x^3+5x^2-4=0\Rightarrow\ x^3+\frac{5}{2}x^2-2=0$

so $\alpha+\beta+\gamma=-\frac{5}{2}$, $\alpha\beta+\alpha\gamma+\beta\gamma=0$ and $\alpha\beta\gamma=2$

$\alpha^3+\beta^3+\gamma^3=(\alpha+\beta+\gamma)^3-3\big((\alpha+\beta+\gamma)(\alpha\beta+\alpha\gamma+\beta\gamma)-\alpha\beta\gamma\big)$

$=\left(-\frac{5}{2}\right)^3-3\left(\left(-\frac{5}{2}\right)(0)-2\right)=-\frac{77}{8}$

41)

$2x^3+7x^2-28x+12=0\Rightarrow x^3+\frac{7}{2}x^2-14x+6=0$

Sum of roots: $\alpha+\beta+\gamma=-\frac{7}{2}$

Sum of pairs of products of roots: $\alpha\beta+\alpha\gamma+\beta\gamma=-14$

Product of roots: $\alpha\beta\gamma=-6$

Sum of new roots: $\alpha\beta+\beta\gamma+\alpha\gamma=-14$

Sum of new pairs of products of roots:

$(\alpha\beta)(\beta\gamma)+(\alpha\beta)(\alpha\gamma)+(\beta\gamma)(\alpha\gamma)=\alpha\beta^2\gamma+\alpha^2\beta\gamma+\alpha\beta\gamma^2=\alpha\beta\gamma(\beta+\alpha+\gamma)$

$=-6\times-\frac{7}{2}=21$

Product of new roots:$(\alpha\beta)(\beta\gamma)(\alpha\gamma)=(\alpha\beta\gamma)^2=(-6)^2=36$

So $z^3+14z^2+21z-36=0$

$\therefore p=14$, $q=21$ and $r=-36$

42) a)

$x^3 + px^2 + qx - 100 = 0$

Product of roots: $\alpha\left(\frac{20}{\alpha}\right)(\alpha - 5) = 100$

$20\alpha - 100 = 100$

$\therefore \alpha = 10$

So the solutions are 10, 2 and 5

42) b)

Sum of roots: $\alpha + \frac{20}{\alpha} + \alpha - 5 = -p$

$10 + \frac{20}{10} + 10 - 5 = -p \therefore p = -17$

Sum of pairs of products of roots: $\alpha\left(\frac{20}{\alpha}\right) + \alpha(\alpha - 5) + \frac{20}{\alpha}(\alpha - 5) = q$

$10\left(\frac{20}{10}\right) + 10(10 - 5) + \frac{20}{10}(10 - 5) = q \therefore q = 80$

$p = -17$ and $q = 80$

43) a)

$x^3 + px^2 + qx - 216 = 0$

Product of roots: $\alpha\left(\frac{36}{\alpha}\right)(\alpha + 4) = 216$

$\alpha + 4 = 6$

$\therefore \alpha = 2$

So the solutions are 2, 18 and 6

43) b)

Sum of roots: $\alpha + \frac{36}{\alpha} + \alpha + 4 = -p$

$2 + 18 + 6 = -p \therefore p = -26$

Sum of pairs of products of roots: $\alpha\left(\frac{36}{\alpha}\right) + \alpha(\alpha + 4) + \frac{36}{\alpha}(\alpha + 4) = q$

$2(18) + 2(6) + 18(6) = q \therefore q = 156$

$p = -26$ and $q = 156$

44) a)

$36x^3 - 444x^2 + px - q = 0 \Rightarrow x^3 - \frac{37}{3}x^2 + \frac{p}{36}x - \frac{q}{36} = 0$

Sum of roots: $\alpha + \frac{1}{\alpha} + \alpha + \frac{1}{\alpha} = \frac{37}{3}$

$\alpha + \frac{1}{\alpha} + \alpha + \frac{1}{\alpha} = \frac{37}{3} \Rightarrow 2\alpha + \frac{2}{\alpha} = \frac{37}{3} \Rightarrow 6\alpha^2 + 6 = 37\alpha \Rightarrow 6\alpha^2 - 37\alpha + 6 = 0$

so $\alpha = 6$ or $\alpha = \frac{1}{6}$

$6 + \frac{1}{6} = \frac{37}{6}$ so the roots are 6, $\frac{1}{6}$ and $\frac{37}{6}$

44) b)

Sum of pairs of products of roots: $\alpha\left(\frac{1}{\alpha}\right) + \alpha\left(\alpha + \frac{1}{\alpha}\right) + \frac{1}{\alpha}\left(\alpha + \frac{1}{\alpha}\right) = \frac{p}{36}$

$6\left(\frac{1}{6}\right) + 6\left(6 + \frac{1}{6}\right) + \frac{1}{6}\left(6 + \frac{1}{6}\right) = \frac{1405}{36} = \frac{p}{36} \therefore p = 1405$

Product of roots: $\alpha\left(\frac{1}{\alpha}\right)\left(\alpha + \frac{1}{\alpha}\right) = \frac{q}{36}$

$6\left(\frac{1}{6}\right)\left(6 + \frac{1}{6}\right) = \frac{37}{6} = \frac{q}{36} \therefore q = 222$

So $p = 1405$ and $q = 222$

45) a)

$2x^3 + 3x^2 - 7x + 2 = 0 \Rightarrow x^3 + \frac{3}{2}x^2 - \frac{7}{2}x + 1 = 0$

so $\alpha + \beta + \gamma = -\frac{3}{2}$, $\alpha\beta + \alpha\gamma + \beta\gamma = -\frac{7}{2}$ and $\alpha\beta\gamma = -1$

$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma}{\alpha\beta\gamma} + \frac{\alpha\gamma}{\alpha\beta\gamma} + \frac{\alpha\beta}{\alpha\beta\gamma} = \frac{\alpha\beta+\alpha\gamma+\beta\gamma}{\alpha\beta\gamma} = \frac{-\frac{7}{2}}{-1} = \frac{7}{2}$

45) b)

New roots: $\frac{1}{\alpha}$, $\frac{1}{\beta}$ and $\frac{1}{\gamma}$

Sum of roots: $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{7}{2}$

Sum of pairs of products of roots: $\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\alpha\gamma} = \frac{\gamma+\alpha+\beta}{\alpha\beta\gamma} = \frac{-\frac{3}{2}}{-1} = \frac{3}{2}$

Product of roots: $\frac{1}{\alpha\beta\gamma} = \frac{1}{-1} = -1$

New equation:

$x^3 - \frac{7}{2}x^2 + \frac{3}{2}x + 1 = 0 \Rightarrow 2x^3 - 7x^2 + 3x + 2 = 0$

46) a)

$3z^3 - 5z^2 + 7z - 4 = 0 \Rightarrow z^3 - \frac{5}{3}z^2 + \frac{7}{3}z - \frac{4}{3} = 0$

so $\alpha + \beta + \gamma = \frac{5}{3}$, $\alpha\beta + \alpha\gamma + \beta\gamma = \frac{7}{3}$ and $\alpha\beta\gamma = \frac{4}{3}$

New roots: $\alpha\beta$, $\alpha\gamma$ and $\beta\gamma$

Sum of roots: $\alpha\beta + \alpha\gamma + \beta\gamma = \frac{7}{3}$

Sum of pairs of products of roots:

$(\alpha\beta)(\alpha\gamma) + (\alpha\beta)(\beta\gamma) + (\alpha\gamma)(\beta\gamma) = \alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2 = \alpha\beta\gamma(\alpha + \beta + \gamma)$

$= \frac{4}{3}\left(\frac{5}{3}\right) = \frac{20}{9}$

Product of roots: $(\alpha\beta)(\alpha\gamma)(\beta\gamma) = (\alpha\beta\gamma)^2 = \left(\frac{4}{3}\right)^2 = \frac{16}{9}$

New equation:

$z^3 - \frac{7}{3}z^2 + \frac{20}{9}z - \frac{16}{9} = 0 \Rightarrow 9z^3 - 21z^2 + 20z - 16 = 0$

46) b)

$3z^3 - 5z^2 + 7z - 4 = 0 \Rightarrow z^3 - \frac{5}{3}z^2 + \frac{7}{3}z - \frac{4}{3} = 0$

so $\alpha + \beta + \gamma = \frac{5}{3}$, $\alpha\beta + \alpha\gamma + \beta\gamma = \frac{7}{3}$ and $\alpha\beta\gamma = \frac{4}{3}$

New roots: $\alpha + \beta$, $\beta + \gamma$ and $\alpha + \gamma$

Sum of roots: $\alpha + \beta + \beta + \gamma + \alpha + \gamma = 2(\alpha + \beta + \gamma) = 2 \times \frac{5}{3} = \frac{10}{3}$

Sum of pairs of products of roots:

$(\alpha + \beta)(\beta + \gamma) + (\alpha + \beta)(\alpha + \gamma) + (\beta + \gamma)(\alpha + \gamma)$

$= \alpha\beta + \alpha\gamma + \beta^2 + \beta\gamma + \alpha^2 + \alpha\gamma + \alpha\beta + \beta\gamma + \alpha\beta + \beta\gamma + \alpha\gamma + \gamma^2$

$= \alpha^2 + \beta^2 + \gamma^2 + 3\alpha\beta + 3\alpha\gamma + 3\beta\gamma$

Since $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$

$(\alpha + \beta)(\beta + \gamma) + (\alpha + \beta)(\alpha + \gamma) + (\beta + \gamma)(\alpha + \gamma)$

$= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) + 3(\alpha\beta + \alpha\gamma + \beta\gamma)$

$= (\alpha + \beta + \gamma)^2 + \alpha\beta + \alpha\gamma + \beta\gamma$

$= \left(\frac{5}{3}\right)^2 + \frac{7}{3} = \frac{46}{9}$

Product of roots:

$(\alpha + \beta)(\alpha + \gamma)(\beta + \gamma) = (\alpha + \beta)(\alpha\beta + \alpha\gamma + \beta\gamma + \gamma^2)$

$= \alpha^2\beta + \alpha^2\gamma + \alpha\beta\gamma + \alpha\gamma^2 + \alpha\beta^2 + \alpha\beta\gamma + \beta^2\gamma + \beta\gamma^2$

$= \alpha^2\beta + \alpha^2\gamma + \alpha\gamma^2 + \alpha\beta^2 + \beta^2\gamma + \beta\gamma^2 + 2\alpha\beta\gamma$

$= (\alpha + \beta + \gamma)(\alpha\beta + \alpha\gamma + \beta\gamma) - \alpha\beta\gamma$

$= \left(\frac{5}{3}\right)\left(\frac{7}{3}\right) - \frac{4}{3} = \frac{23}{9}$

New equation:

$z^3 - \frac{10}{3}z^2 + \frac{46}{9}z - \frac{23}{9} = 0 \Rightarrow 9z^3 - 30z^2 + 46z - 23 = 0$

47)

$3x^3 + 4x^2 - 8x + k = 0 \Rightarrow x^3 + \frac{4}{3}x^2 - \frac{8}{3}x + \frac{k}{3} = 0$

so $\alpha + \beta + \gamma = -\frac{4}{3}$, $\alpha\beta + \alpha\gamma + \beta\gamma = -\frac{8}{3}$ and $\alpha\beta\gamma = -\frac{k}{3}$

$(\alpha + \beta)(\alpha + \gamma)(\beta + \gamma) = (\alpha + \beta)(\alpha\beta + \alpha\gamma + \beta\gamma + \gamma^2)$

$= \alpha^2\beta + \alpha^2\gamma + \alpha\beta\gamma + \alpha\gamma^2 + \alpha\beta^2 + \alpha\beta\gamma + \beta^2\gamma + \beta\gamma^2$

$= \alpha^2\beta + \alpha^2\gamma + \alpha\gamma^2 + \alpha\beta^2 + \beta^2\gamma + \beta\gamma^2 + 2\alpha\beta\gamma$

$= (\alpha + \beta + \gamma)(\alpha\beta + \alpha\gamma + \beta\gamma) - \alpha\beta\gamma = \left(-\frac{4}{3}\right)\left(-\frac{8}{3}\right) + \frac{k}{3} = \frac{32}{9} + \frac{k}{3}$

48)

$px^3 + 11x^2 - 106x + q = 0 \Rightarrow x^3 + \frac{11}{p}x^2 - \frac{106}{p}x + \frac{q}{p} = 0$

so $\alpha + \beta + \gamma = -\frac{11}{p}$, $\alpha\beta + \alpha\gamma + \beta\gamma = -\frac{106}{p}$ and $\alpha\beta\gamma = -\frac{q}{p}$

$20w^3 + 93w^2 - rw - 685 = 0 \Rightarrow w^3 + \frac{93}{20}w^2 - \frac{r}{20}w - \frac{137}{4} = 0$

Sum of roots: $-\frac{93}{20}$

$3\alpha - 1 + 3\beta - 1 + 3\gamma - 1 = -\frac{93}{20}$

$3(\alpha + \beta + \gamma) - 3 = -\frac{93}{20}$

$3\left(-\frac{11}{p}\right) - 3 = -\frac{93}{20} \Rightarrow \frac{11}{p} = \frac{11}{20} \therefore p = 20$

Sum of pairs of products of roots: $-\frac{r}{20}$

$(3\alpha - 1)(3\beta - 1) + (3\alpha - 1)(3\gamma - 1) + (3\beta - 1)(3\gamma - 1)$

$= 9\alpha\beta - 3\alpha - 3\beta + 1 + 9\alpha\gamma - 3\alpha - 3\gamma + 1 + 9\beta\gamma - 3\beta - 3\gamma + 1$

$= 9\alpha\beta + 9\alpha\gamma + 9\beta\gamma - 6\alpha - 6\beta - 6\gamma + 3$

$= 9(\alpha\beta + \alpha\gamma + \beta\gamma) - 6(\alpha + \beta + \gamma) + 3$

So that

$9(\alpha\beta + \alpha\gamma + \beta\gamma) - 6(\alpha + \beta + \gamma) + 3 = -\frac{r}{20}$

$9\left(-\frac{106}{20}\right) - 6\left(-\frac{11}{20}\right) + 3 = -\frac{r}{20} \Rightarrow -\frac{207}{5} = -\frac{r}{20} \therefore r = 828$

Product of roots: $\frac{137}{4}$

$(3\alpha - 1)(3\beta - 1)(3\gamma - 1) = 27\alpha\beta\gamma - 9(\alpha\beta + \alpha\gamma + \beta\gamma) + 3(\alpha + \beta + \gamma) - 1$

So

$27\alpha\beta\gamma - 9(\alpha\beta + \alpha\gamma + \beta\gamma) + 3(\alpha + \beta + \gamma) - 1 = \frac{137}{4}$

$27\left(-\frac{q}{20}\right) - 9\left(-\frac{106}{20}\right) + 3\left(-\frac{11}{20}\right) - 1 = \frac{137}{4} \Rightarrow 27\left(-\frac{q}{20}\right) = -\frac{54}{5} \therefore q = 8$

$\therefore p = 20$, $q = 8$ and $r = 828$

49) a)

$x^3 + 10x^2 + kx - 126 = 0$

Sum of roots: $\alpha + \beta + \gamma = -10$

Sum of pairs of products of roots: $\alpha\beta + \alpha\gamma + \beta\gamma = k$

Product of roots: $\alpha\beta\gamma = 94$

$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$

$94 = (-10)^2 - 2(k)$

$\therefore k = 3$

49) b)

Sum of roots: $2\alpha + 2\beta + 2\gamma = 2(\alpha + \beta + \gamma) = -20$

Sum of pairs of products of roots: $4\alpha\beta + 4\alpha\gamma + 4\beta\gamma = 4(\alpha\beta + \alpha\gamma + \beta\gamma) = 4 \times 3 = 12$

Product of roots: $8\alpha\beta\gamma = 8 \times 126 = 1008$

$x^3 + 20x^2 + 12x - 1008 = 0$

50) a)

$z^3 - pz^2 + 4z + q = 0$

Sum of roots: $\alpha + \beta + \gamma = p$

Sum of pairs of products of roots: $\alpha\beta + \alpha\gamma + \beta\gamma = 4$

$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$

$17 = p^2 - 2 \times 4$ so $p = \pm 5$

Since p is positive, $p = 5$

50) b)

Product of roots: $\alpha\beta\gamma = -q$

$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma}{\alpha\beta\gamma} + \frac{\alpha\gamma}{\alpha\beta\gamma} + \frac{\alpha\beta}{\alpha\beta\gamma} = \frac{\alpha\beta+\alpha\gamma+\beta\gamma}{\alpha\beta\gamma} = \frac{4}{-q}$

$-\frac{2}{3} = -\frac{4}{q}$

$\therefore q = 6$

50) c)

$(\alpha + 1)(\beta + 1)(\gamma + 1) = \alpha\beta\gamma + \alpha\beta + \alpha\gamma + \alpha + \beta\gamma + \beta + \gamma + 1$

$= -6 + 4 + 5 + 1 = 4$

51) a)

For f the sum of roots is p

For g the product of roots is 4

$p = 4$

51) b)

For g:

$3(1-\mathrm{i})^3 - 12(1-\mathrm{i})^2 + q(1-\mathrm{i}) - 12 = 0$

$3(-2-2\mathrm{i}) - 12(-2\mathrm{i}) + q(1-\mathrm{i}) - 12 = 0$

$-6 - 6\mathrm{i} + 24\mathrm{i} + q - q\mathrm{i} - 12 = 0$

$-18 + q + 18\mathrm{i} - q\mathrm{i} = 0$

$\therefore q = 18$

52) a)

$3x^3 + kx^2 + 85x - 75 = 0 \Rightarrow x^3 + \frac{k}{3}x^2 + \frac{85}{3}x - 25 = 0$

Product of roots: $\alpha\beta\left(\frac{\alpha}{\beta}\right) = 25$

$\alpha^2 = 25$ so $\alpha = \pm 5$

Sum of pairs of products of roots: $\alpha\beta + \alpha\left(\frac{\alpha}{\beta}\right) + \beta\left(\frac{\alpha}{\beta}\right) = \frac{85}{3}$

$5\beta + \frac{25}{\beta} + 5 = \frac{85}{3} \Rightarrow 3\beta^2 + 15 - 14\beta = 0 \therefore \beta = 3$ or $\beta = \frac{5}{3}$

$-5\beta + \frac{25}{\beta} - 5 = \frac{85}{3} \Rightarrow 3\beta^2 - 15 + 20\beta = 0 \therefore \beta = -\frac{10}{3} - \frac{\sqrt{145}}{3}$ or $\beta = -\frac{10}{3} + \frac{\sqrt{145}}{3}$

So the possible solutions are:

$5, 3$ and $\frac{5}{3}$ or $-5, -\frac{10}{3} - \frac{\sqrt{145}}{3}$ and $-\frac{10}{3} + \frac{\sqrt{145}}{3}$

52) b)

Sum of roots: $-\frac{k}{3}$

$5 + 3 + \frac{5}{3} = -\frac{k}{3} \Rightarrow k = -29$

$-5 - \frac{10}{3} - \frac{\sqrt{145}}{3} - \frac{10}{3} + \frac{\sqrt{145}}{3} = -\frac{k}{3} \Rightarrow k = 35$

$k = -29$ or $k = 35$

53)

$x^3 + kx^2 + 7 = 0$

Sum of roots: $\alpha + \beta + \gamma = -k$

Sum of pairs of products of roots: $\alpha\beta + \alpha\gamma + \beta\gamma = 0$

Product of roots: $\alpha\beta\gamma = -7$

$8x^3 - 32x^2 + px + q = 0 \Rightarrow x^3 - 4x^2 + \frac{p}{8}x + \frac{q}{8} = 0$

Sum of roots: $\alpha\beta + \gamma + \beta\gamma + \alpha + \alpha\gamma + \beta = 4$

$-k = 4 \therefore k = -4$

54)

$z^3 - az^2 - 13z + b = 0$

Sum of roots: $\alpha + \beta - 4 = a$

So $\alpha + \beta = a + 4$

Product of roots: $-4\alpha\beta = -b$

So $\alpha\beta = \frac{b}{4}$

Sum of pairs of products of roots: $\alpha\beta - 4\alpha - 4\beta = -13$

$\alpha\beta - 4(\alpha + \beta) = -13 \Rightarrow \frac{b}{4} - 4(a + 4) = -13$

$\frac{b}{4} - 4a - 16 = -13$

$b - 16a = 12$

$\therefore b = 12 + 16a$

55) a)

$z^3 - 4z^2 + kz - 8 = 0$

Sum of roots: $\alpha + \beta + \alpha - \beta + \frac{8}{\alpha} = 4$

$2\alpha + \frac{8}{\alpha} = 4 \Rightarrow 2\alpha^2 - 4\alpha + 8 = 0$ so $\alpha = 1 \pm \sqrt{3}\mathrm{i}$

Product of roots: $\alpha\beta\left(\alpha - \beta + \frac{8}{\alpha}\right) = 8$

$(1 + \sqrt{3}\mathrm{i})\beta\left(1 + \sqrt{3}\mathrm{i} - \beta + \frac{8}{1+\sqrt{3}\mathrm{i}}\right) = 8$

$\beta\left(\left(1 + \sqrt{3}\mathrm{i}\right)^2 - \beta\left(1 + \sqrt{3}\mathrm{i}\right) + 8\right) = 8$

$\beta\left(1 + 2\sqrt{3}\mathrm{i} - 3 - \beta - \sqrt{3}\beta\mathrm{i} + 8\right) = 8$

$\beta + 2\sqrt{3}\beta\mathrm{i} - 3\beta - \beta^2 - \sqrt{3}\beta^2\mathrm{i} + 8\beta - 8 = 0$

$(-\beta^2 + 6\beta - 8) + \left(2\sqrt{3}\beta - \sqrt{3}\beta^2\right)\mathrm{i} = 0$

Comparing real: $-\beta^2 + 6\beta - 8 = 0 \Rightarrow \beta = 2$ or $\beta = 4$

Comparing imaginary: $\beta\left(2\sqrt{3} - \sqrt{3}\beta\right) = 0 \Rightarrow \beta = 0$ or $\beta = 2$

So $\beta = 2$

Since the coefficients are real, the solutions are 2, $1 + \sqrt{3}\mathrm{i}$ and $1 - \sqrt{3}\mathrm{i}$

55) b)

Sum of pairs of products of roots:

$2\left(1 + \sqrt{3}\mathrm{i}\right) + 2\left(1 - \sqrt{3}\mathrm{i}\right) + \left(1 + \sqrt{3}\mathrm{i}\right)\left(1 - \sqrt{3}\mathrm{i}\right) = k$

$2 + 2\sqrt{3}\mathrm{i} + 2 - 2\sqrt{3}\mathrm{i} + 1 + 3 = k$

$\therefore k = 8$

56)

$x^3 - (p+4)x^2 + 3px - 9p = 0$

Sum of roots: $\alpha + \beta + \gamma = p + 4$

Sum of pairs of products of roots: $\alpha\beta + \alpha\gamma + \beta\gamma = 3p$

Product of roots: $\alpha\beta\gamma = 9p$

$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$

$= (p+4)^2 - 2(3p)$

$= p^2 + 2p + 16$ as req.

57)

$x^3 - (p+6)x^2 + (5p+4)x - (7p+3) = 0$

Sum of roots: $\alpha + \beta + \gamma = p + 6$

Sum of pairs of products of roots: $\alpha\beta + \alpha\gamma + \beta\gamma = 5p + 4$

Product of roots: $\alpha\beta\gamma = 7p + 3$

$\alpha^2\beta^2 + \beta^2\gamma^2 + \alpha^2\gamma^2 = (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)$

$= (5p+4)^2 - 2(7p+3)(p+6)$

$= 25p^2 + 40p + 16 - 2(7p^2 + 45p + 18)$

$= 25p^2 + 40p + 16 - 14p^2 - 90p - 36$

$= 11p^2 - 50p - 20$ as req.

58)

$x^3 - (p+3)x^2 + 4px - 12p = 0$

Sum of roots: $\alpha + \beta + \gamma = p + 3$

Sum of pairs of products of roots: $\alpha\beta + \alpha\gamma + \beta\gamma = 4p$

Product of roots: $\alpha\beta\gamma = 12p$

$(\alpha + \beta + \gamma)^3 = (\alpha + \beta + \gamma)(\alpha^2 + \beta^2 + \gamma^2 + 2\alpha\beta + 2\alpha\gamma + 2\beta\gamma)$

$= \alpha^3 + \alpha\beta^2 + \alpha\gamma^2 + 2\alpha^2\beta + 2\alpha^2\gamma + 2\alpha\beta\gamma + \alpha^2\beta + \beta^3 + \beta\gamma^2 + 2\alpha\beta^2 + 2\alpha\beta\gamma +$
$2\beta^2\gamma + \alpha^2\gamma + \beta^2\gamma + \gamma^3 + 2\alpha\beta\gamma + 2\alpha\gamma^2 + 2\beta\gamma^2$

$= \alpha^3 + \beta^3 + \gamma^3 + 3\alpha\beta^2 + 3\alpha\gamma^2 + 3\alpha^2\beta + 3\alpha^2\gamma + 6\alpha\beta\gamma + 3\beta\gamma^2 + 3\beta^2\gamma$

$= \alpha^3 + \beta^3 + \gamma^3 + 3(\alpha\beta^2 + \alpha\gamma^2 + \alpha^2\beta + \alpha^2\gamma + \beta\gamma^2 + \beta^2\gamma + 2\alpha\beta\gamma)$

But given

$\alpha^2\beta + \alpha^2\gamma + \alpha\gamma^2 + \alpha\beta^2 + \beta^2\gamma + \beta\gamma^2 + 2\alpha\beta\gamma = (\alpha + \beta + \gamma)(\alpha\beta + \alpha\gamma + \beta\gamma) - \alpha\beta\gamma$

This gives

$(\alpha + \beta + \gamma)^3 = \alpha^3 + \beta^3 + \gamma^3 + 3\big((\alpha + \beta + \gamma)(\alpha\beta + \alpha\gamma + \beta\gamma) - \alpha\beta\gamma\big)$

$\alpha^3 + \beta^3 + \gamma^3 = (\alpha + \beta + \gamma)^3 - 3\big((\alpha + \beta + \gamma)(\alpha\beta + \alpha\gamma + \beta\gamma) - \alpha\beta\gamma\big)$

And so

$\alpha^3 + \beta^3 + \gamma^3 = (p+3)^3 - 3\big((p+3)(4p) - 12p\big)$

$= p^3 + 9p^2 + 27p + 27 - 3(4p^2 + 12p - 12p)$

$= p^3 + 9p^2 + 27p + 27 - 12p^2$

$= p^3 - 3p^2 + 27p + 27$

59)

$x^3 + px^2 + qx + r = 0$

Sum of roots: $\alpha + \alpha + \alpha = -p$

$3\alpha = -p$

Sum of pairs of products of roots: $\alpha\alpha + \alpha\alpha + \alpha\alpha = q$

$3\alpha^2 = q$

Product of roots: $\alpha\alpha\alpha = -r$

$\alpha^3 = -r$

$\alpha = -\frac{p}{3}$ and $\alpha^2 = \frac{q}{3}$

So $-\frac{p}{3} \times \frac{q}{3} = -r$

$p = \frac{9r}{q}$

60)

$x^3 - 49x^2 - 1029x + k = 0$

Let the roots be p, pr and pr^2

Sum of roots: $p + pr + pr^2 = 49$

$p(1 + r + r^2) = 49$

Sum of pairs of products of roots: $p(pr) + p(pr^2) + (pr)(pr^2) = -1029$

$p^2r + p^2r^2 + p^2r^3 = -1029$

$p^2r(1 + r + r^2) = -1029$

Product of roots: $p(pr)(pr^2) = -k$

$(pr)^3 = k$

Substituting $p(1 + r + r^2) = 49$ into $p^2r(1 + r + r^2) = -1029$ gives

$49pr = -1029$

$\therefore pr = -21$

So $-k = (-21)^3 = -9261 \therefore k = 9261$

61)

$ax^3 + bx^2 + cx + d = 0 \Rightarrow x^3 + \frac{b}{a}x^2 + \frac{c}{a}x + \frac{d}{a} = 0$

Let the roots be p, pr and pr^2

Sum of roots: $p + pr + pr^2 = -\frac{b}{a}$

$p(1 + r + r^2) = -\frac{b}{a}$

Sum of pairs of products of roots: $p(pr) + p(pr^2) + (pr)(pr^2) = \frac{c}{a}$

$p^2r + p^2r^2 + p^2r^3 = \frac{c}{a}$

$p^2r(1 + r + r^2) = \frac{c}{a}$

Product of roots: $p(pr)(pr^2) = -\frac{d}{a}$

$p^3r^3 = -\frac{d}{a}$

Substituting $p(1 + r + r^2) = -\frac{b}{a}$ into $p^2r(1 + r + r^2) = \frac{c}{a}$ gives:

$pr\left(-\frac{b}{a}\right) = \frac{c}{a}$

Cubing both sides gives:

$(pr)^3\left(-\frac{b}{a}\right)^3 = \left(\frac{c}{a}\right)^3$

$\left(-\frac{d}{a}\right)\left(-\frac{b}{a}\right)^3 = \left(\frac{c}{a}\right)^3$

$\frac{b^3d}{a^4} = \frac{c^3}{a^3}$

$\Rightarrow b^3d = ac^3 \Rightarrow ac^3 - db^3 = 0$ as req.

62)

$2x^3 + 6x^2 - 26x - k = 0 \Rightarrow x^3 + 3x^2 - 13x - \frac{k}{2} = 0$

Let the roots be $a - d, a$ and $a + d$

Sum of roots: $a - d + a + a + d = -3$

$3a = -3$

So $a = -1$

Sum of pairs of products of roots: $-(a - d) + a(a + d) + (a - d)(a + d) = -13$

$-(-1 - d) - (-1 + d) + (-1 - d)(-1 + d) = -13$

$1 + d + 1 - d + 1 - d^2 = -13$

$-d^2 + 3 = -13$

$d^2 = 16$

So $d = \pm 4$

Product of roots: $(a - d)a(a + d) = \frac{k}{2}$

$-1(-1 - 4)(-1 + 4) = \frac{k}{2}$

$15 = \frac{k}{2}$

$\therefore k = 30$

63)

$px^3 + qx^2 + rx + s = 0 \Rightarrow x^3 + \frac{q}{p}x^2 + \frac{r}{p}x + \frac{s}{p} = 0$

Let the roots be $\alpha - d$, α and $\alpha + d$

Sum of roots: $\alpha - d + \alpha + \alpha + d = -\frac{q}{p}$

$3\alpha = -\frac{q}{p}$

Sum of pairs of products of roots: $\alpha(\alpha - d) + \alpha(\alpha + d) + (\alpha - d)(\alpha + d) = \frac{r}{p}$

$\alpha^2 - d\alpha + \alpha^2 + d\alpha + \alpha^2 - d^2 = \frac{r}{p}$

$2\alpha^2 + \alpha^2 - d^2 = \frac{r}{p}$

Product of roots: $\alpha(\alpha - d)(\alpha + d) = -\frac{s}{p}$

$\alpha(\alpha^2 - d^2) = -\frac{s}{p}$

Substituting $\alpha(\alpha^2 - d^2) = -\frac{s}{p}$ into $2\alpha^2 + \alpha^2 - d^2 = \frac{r}{p}$ gives:

$2\alpha^2 - \frac{s}{p\alpha} = \frac{r}{p}$

Since $\alpha = -\frac{q}{3p}$

$2\left(-\frac{q}{3p}\right)^2 - \frac{s}{p\left(-\frac{q}{3p}\right)} = \frac{r}{p}$

$\frac{2q^2}{9p^2} + \frac{3s}{q} = \frac{r}{p}$

$2q^3 + 27p^2s = 9pqr$ as req.

64)

$x^3 + px^2 + qx + r = 0$

Sum of roots: $\alpha + \beta + \gamma = -p$

Sum of pairs of products of roots: $\alpha\beta + \alpha\gamma + \beta\gamma = q$

Product of roots: $\alpha\beta\gamma = -r$

$q^2 - 2pr = (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)$

But since

$(\alpha\beta + \alpha\gamma + \beta\gamma)^2 = (\alpha\beta)^2 + (\alpha\gamma)^2 + (\beta\gamma)^2 + 2\alpha^2\beta\gamma + 2\alpha\beta^2\gamma + 2\alpha\beta\gamma^2$

$= (\alpha\beta)^2 + (\alpha\gamma)^2 + (\beta\gamma)^2 - 2(\alpha + \beta + \gamma)(\alpha\beta\gamma)$

So that

$q^2 - 2pr = (\alpha\beta)^2 + (\alpha\gamma)^2 + (\beta\gamma)^2 + 2(\alpha + \beta + \gamma)(\alpha\beta\gamma) - 2\alpha\beta\gamma(\alpha + \beta + \gamma)$

$= (\alpha\beta)^2 + (\alpha\gamma)^2 + (\beta\gamma)^2 \geq 0$

$\Rightarrow q^2 \geq 2pr$ as req.

65) a)

$\sum \alpha = -5$, $\sum \alpha\beta = -3$, $\sum \alpha\beta\gamma = -6$ and $\sum \alpha\beta\gamma\delta = -2$

65) b)

$\sum \alpha = 4$, $\sum \alpha\beta = 4$, $\sum \alpha\beta\gamma = 1$ and $\sum \alpha\beta\gamma\delta = 8$

65) c)

$\sum \alpha = -\frac{5}{2}$, $\sum \alpha\beta = 3$, $\sum \alpha\beta\gamma = -\frac{11}{2}$ and $\sum \alpha\beta\gamma\delta = \frac{1}{2}$

65) d)

$\sum \alpha = -1$, $\sum \alpha\beta = -\frac{7}{4}$, $\sum \alpha\beta\gamma = 5$ and $\sum \alpha\beta\gamma\delta = \frac{1}{2}$

65) e)

$x^4 + 7 - x^3 + 4x^2 = 0$

$\sum \alpha = 1$, $\sum \alpha\beta = 4$, $\sum \alpha\beta\gamma = 0$ and $\sum \alpha\beta\gamma\delta = 7$

66) a)

$8z^4 + z^3 - 3z^2 + 5z - 7 = 0 \Rightarrow z^4 + \frac{1}{8}z^3 - \frac{3}{8}z^2 + \frac{5}{8}z - \frac{7}{8} = 0$

$\alpha + \beta + \gamma + \delta = -\frac{1}{8}$

66) b)

$8z^4 + z^3 - 3z^2 + 5z - 7 = 0 \Rightarrow z^4 + \frac{1}{8}z^3 - \frac{3}{8}z^2 + \frac{5}{8}z - \frac{7}{8} = 0$

$\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = -\frac{3}{8}$

66) c)

$8z^4 + z^3 - 3z^2 + 5z - 7 = 0 \Rightarrow z^4 + \frac{1}{8}z^3 - \frac{3}{8}z^2 + \frac{5}{8}z - \frac{7}{8} = 0$

$\alpha\beta\gamma + \alpha\gamma\delta + \alpha\beta\delta + \beta\gamma\delta = -\frac{5}{8}$

66) d)

$8z^4 + z^3 - 3z^2 + 5z - 7 = 0 \Rightarrow z^4 + \frac{1}{8}z^3 - \frac{3}{8}z^2 + \frac{5}{8}z - \frac{7}{8} = 0$

$\alpha\beta\gamma\delta = -\frac{7}{8}$

66) e)

$8z^4 + z^3 - 3z^2 + 5z - 7 = 0 \Rightarrow z^4 + \frac{1}{8}z^3 - \frac{3}{8}z^2 + \frac{5}{8}z - \frac{7}{8} = 0$

$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta} = \frac{\beta\gamma\delta}{\alpha\beta\gamma\delta} + \frac{\alpha\gamma\delta}{\alpha\beta\gamma\delta} + \frac{\alpha\beta\delta}{\alpha\beta\gamma\delta} + \frac{\alpha\beta\gamma}{\alpha\beta\gamma\delta} = \frac{\beta\gamma\delta+\alpha\gamma\delta+\alpha\beta\delta+\alpha\beta\gamma}{\alpha\beta\gamma\delta} = \frac{-\frac{5}{8}}{-\frac{7}{8}} = \frac{5}{7}$

66) f)

$8z^4 + z^3 - 3z^2 + 5z - 7 = 0 \Rightarrow z^4 + \frac{1}{8}z^3 - \frac{3}{8}z^2 + \frac{5}{8}z - \frac{7}{8} = 0$

$\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = (\alpha + \beta + \gamma + \delta)^2 - 2(\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta)$

$= \left(-\frac{1}{8}\right)^2 - 2\left(-\frac{3}{8}\right) = \frac{49}{64}$

66) g)

$8z^4 + z^3 - 3z^2 + 5z - 7 = 0 \Rightarrow z^4 + \frac{1}{8}z^3 - \frac{3}{8}z^2 + \frac{5}{8}z - \frac{7}{8} = 0$

$\alpha^2\beta^2\gamma\delta + \alpha^2\beta\gamma^2\delta + \alpha^2\beta\gamma\delta^2 + \alpha\beta^2\gamma^2\delta + \alpha\beta^2\gamma\delta^2 + \alpha\beta\gamma^2\delta^2$

$= (\alpha\beta\gamma\delta)(\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta)$

$= \left(-\frac{7}{8}\right)\left(-\frac{3}{8}\right) = \frac{21}{64}$

66) h)

$8z^4 + z^3 - 3z^2 + 5z - 7 = 0 \Rightarrow z^4 + \frac{1}{8}z^3 - \frac{3}{8}z^2 + \frac{5}{8}z - \frac{7}{8} = 0$

$\frac{2}{\alpha\beta\gamma} + \frac{2}{\alpha\beta\delta} + \frac{2}{\alpha\gamma\delta} + \frac{2}{\beta\gamma\delta} = 2\left[\frac{\delta}{\alpha\beta\gamma\delta} + \frac{\gamma}{\alpha\beta\gamma\delta} + \frac{\beta}{\alpha\beta\gamma\delta} + \frac{\alpha}{\alpha\beta\gamma\delta}\right]$

$= 2\left(\frac{\alpha+\beta+\gamma+\delta}{\alpha\beta\gamma\delta}\right) = 2\left(\frac{-\frac{1}{8}}{-\frac{7}{8}}\right) = \frac{2}{7}$

67) a)

$12z^4 + 3z^3 - 2z^2 - 4z + 3 = 0 \Rightarrow z^4 + \frac{1}{4}z^3 - \frac{1}{6}z^2 - \frac{1}{3}z + \frac{1}{4} = 0$

$\alpha + \beta + \gamma + \delta = -\frac{1}{4}$

67) b)

$12z^4 + 3z^3 - 2z^2 - 4z + 3 = 0 \Rightarrow z^4 + \frac{1}{4}z^3 - \frac{1}{6}z^2 - \frac{1}{3}z + \frac{1}{4} = 0$

$\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = -\frac{1}{6}$

67) c)

$12z^4 + 3z^3 - 2z^2 - 4z + 3 = 0 \Rightarrow z^4 + \frac{1}{4}z^3 - \frac{1}{6}z^2 - \frac{1}{3}z + \frac{1}{4} = 0$

$\alpha\beta\gamma + \alpha\gamma\delta + \alpha\beta\delta + \beta\gamma\delta = \frac{1}{3}$

67) d)

$12z^4 + 3z^3 - 2z^2 - 4z + 3 = 0 \Rightarrow z^4 + \frac{1}{4}z^3 - \frac{1}{6}z^2 - \frac{1}{3}z + \frac{1}{4} = 0$

$\alpha\beta\gamma\delta = \frac{1}{4}$

67) e)

$12z^4 + 3z^3 - 2z^2 - 4z + 3 = 0 \Rightarrow z^4 + \frac{1}{4}z^3 - \frac{1}{6}z^2 - \frac{1}{3}z + \frac{1}{4} = 0$

$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta} = \frac{\beta\gamma\delta}{\alpha\beta\gamma\delta} + \frac{\alpha\gamma\delta}{\alpha\beta\gamma\delta} + \frac{\alpha\beta\delta}{\alpha\beta\gamma\delta} + \frac{\alpha\beta\gamma}{\alpha\beta\gamma\delta} = \frac{\beta\gamma\delta+\alpha\gamma\delta+\alpha\beta\delta+\alpha\beta\gamma}{\alpha\beta\gamma\delta} = \frac{\frac{1}{3}}{\frac{1}{4}} = \frac{4}{3}$

67) f)

$12z^4 + 3z^3 - 2z^2 - 4z + 3 = 0 \Rightarrow z^4 + \frac{1}{4}z^3 - \frac{1}{6}z^2 - \frac{1}{3}z + \frac{1}{4} = 0$

$\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = (\alpha + \beta + \gamma + \delta)^2 - 2(\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta)$

$= \left(-\frac{1}{4}\right)^2 - 2\left(-\frac{1}{6}\right) = \frac{19}{48}$

67) g)

$12z^4 + 3z^3 - 2z^2 - 4z + 3 = 0 \Rightarrow z^4 + \frac{1}{4}z^3 - \frac{1}{6}z^2 - \frac{1}{3}z + \frac{1}{4} = 0$

$\alpha^2\beta^2\gamma\delta + \alpha^2\beta\gamma^2\delta + \alpha^2\beta\gamma\delta^2 + \alpha\beta^2\gamma^2\delta + \alpha\beta^2\gamma\delta^2 + \alpha\beta\gamma^2\delta^2$

$= (\alpha\beta\gamma\delta)(\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta)$

$= \left(\frac{1}{4}\right)\left(-\frac{1}{6}\right) = -\frac{1}{24}$

67) h)

$12z^4 + 3z^3 - 2z^2 - 4z + 3 = 0 \Rightarrow z^4 + \frac{1}{4}z^3 - \frac{1}{6}z^2 - \frac{1}{3}z + \frac{1}{4} = 0$

$\frac{2}{\alpha\beta\gamma} + \frac{2}{\alpha\beta\delta} + \frac{2}{\alpha\gamma\delta} + \frac{2}{\beta\gamma\delta} = 2\left[\frac{\delta}{\alpha\beta\gamma\delta} + \frac{\gamma}{\alpha\beta\gamma\delta} + \frac{\beta}{\alpha\beta\gamma\delta} + \frac{\alpha}{\alpha\beta\gamma\delta}\right]$

$= 2\left(\frac{\alpha+\beta+\gamma+\delta}{\alpha\beta\gamma\delta}\right) = 2\left(\frac{-\frac{1}{4}}{\frac{1}{4}}\right) = -2$

68) a)

$2z^4 + 3z^3 - 2z + 3 = 0 \Rightarrow z^4 + \frac{3}{2}z^3 - z + \frac{3}{2} = 0$

$\sum \alpha = \alpha + \beta + \gamma + \delta = -\frac{3}{2}$

$\sum \alpha\beta = \alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = 0$

$\sum \alpha\beta\gamma = \alpha\beta\gamma + \alpha\gamma\delta + \alpha\beta\delta + \beta\gamma\delta = 1$

$\sum \alpha\beta\gamma\delta = \alpha\beta\gamma\delta = \frac{3}{2}$

$\frac{3}{\alpha} + \frac{3}{\beta} + \frac{3}{\gamma} + \frac{3}{\delta} = 3\left[\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta}\right] = 3\left[\frac{\beta\gamma\delta}{\alpha\beta\gamma\delta} + \frac{\alpha\gamma\delta}{\alpha\beta\gamma\delta} + \frac{\alpha\beta\delta}{\alpha\beta\gamma\delta} + \frac{\alpha\beta\gamma}{\alpha\beta\gamma\delta}\right] = 3\left[\frac{\beta\gamma\delta+\alpha\gamma\delta+\alpha\beta\delta+\alpha\beta\gamma}{\alpha\beta\gamma\delta}\right]$

$= 3\left[\frac{1}{\frac{3}{2}}\right] = 2$

68) b)

$2z^4 + 3z^3 - 2z + 3 = 0 \Rightarrow z^4 + \frac{3}{2}z^3 - z + \frac{3}{2} = 0$

$\sum \alpha = \alpha + \beta + \gamma + \delta = -\frac{3}{2}$

$\sum \alpha\beta = \alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = 0$

$\sum \alpha\beta\gamma = \alpha\beta\gamma + \alpha\gamma\delta + \alpha\beta\delta + \beta\gamma\delta = 1$

$\sum \alpha\beta\gamma\delta = \alpha\beta\gamma\delta = \frac{3}{2}$

$\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = (\alpha + \beta + \gamma + \delta)^2 - 2(\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta)$

$= \left(-\frac{3}{2}\right)^2 - 2(0) = \frac{9}{4}$

68) c)

$(1 - \alpha)(1 - \beta)(1 - \gamma)(1 - \delta)$

$= \alpha\beta\gamma\delta - (\alpha\beta\gamma + \alpha\gamma\delta + \alpha\beta\delta + \beta\gamma\delta) + (\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta) -$
$(\alpha + \beta + \gamma + \delta) + 1$

$= \frac{3}{2} - 1 - 0 - \left(-\frac{3}{2}\right) + 1 = 3$

69) a)

$6x^4 - 35x^3 + 62x^2 - px + q = 0 \Rightarrow x^4 - \frac{35}{6}x^3 + \frac{31}{3}x^2 - \frac{p}{6}x + \frac{q}{6} = 0$

$\sum \alpha\beta\gamma\delta = \alpha \times \frac{1}{\alpha} \times \beta \times \frac{1}{\beta} = 1 = \frac{q}{6}$

$\therefore q = 6$

69) b)

$\sum \alpha = \frac{35}{6}, \ \sum \alpha\beta\gamma = \frac{p}{6}$

Since

$\sum \alpha = \alpha + \frac{1}{\alpha} + \beta + \frac{1}{\beta} = \frac{35}{6}$

$\sum \alpha\beta\gamma = \alpha\beta\left(\frac{1}{\alpha}\right) + \alpha\left(\frac{1}{\alpha}\right)\left(\frac{1}{\beta}\right) + \alpha\beta\left(\frac{1}{\beta}\right) + \beta\left(\frac{1}{\alpha}\right)\left(\frac{1}{\beta}\right) = \beta + \frac{1}{\beta} + \alpha + \frac{1}{\alpha} = \frac{p}{6}$

$\therefore p = 35$

70) a)

$2x^4 + 3x^3 + px^2 + qx + 882 = 0 \Rightarrow x^4 + \frac{3}{2}x^3 + \frac{p}{2}x^2 + \frac{q}{2}x + 441 = 0$

$\sum \alpha = -\frac{3}{2}, \sum \alpha\beta = \frac{p}{2}, \sum \alpha\beta\gamma = -\frac{q}{2}, \sum \alpha\beta\gamma\delta = 441$

So

$\alpha + 2\alpha + \beta + \frac{\beta}{2} = -\frac{3}{2} \Rightarrow 3\alpha + \frac{3}{2}\beta = -\frac{3}{2} \Rightarrow 2\alpha + \beta = -1 \Rightarrow \beta = -1 - 2\alpha$

$\alpha \times 2\alpha \times \beta \times \frac{\beta}{2} = \alpha^2\beta^2 = 441$

Solving simultaneously:

$\alpha^2(-1 - 2\alpha)^2 = 441$

$\alpha^2(4\alpha^2 + 4\alpha + 1) = 441$

$4\alpha^4 + 4\alpha^3 + \alpha^2 - 441 = 0$

$(\alpha - 3)(2\alpha + 7)(2\alpha^2 + \alpha + 21) = 0$

Real solutions are: $\alpha = -\frac{7}{2}$ or $\alpha = 3$

$\alpha = -\frac{7}{2}, \beta = 6$ or $\alpha = 3, \beta = -7$

So the roots are $-\frac{7}{2}, -7, 6$ and 3

70) b)

$(2x + 7)(x - 6)(x - 3)(x + 7) = 0$

$(2x^2 - 5x - 42)(x^2 + 4x - 21) = 0$

$2x^4 + 8x^3 - 42x^2 - 5x^3 - 20x^2 + 105x - 42x^2 - 168x + 882 = 0$

$2x^4 + 3x^3 - 104x^2 - 63x + 882 = 0$

$\therefore p = -104$ and $q = -63$

71) a)

$9x^4 - 150x^3 + px^2 - qx + 63 = 0 \Rightarrow x^4 - \frac{150}{9}x^3 + \frac{p}{9}x^2 - \frac{q}{9}x + 7 = 0$

Let $\delta = \gamma$ so that the roots are α, β, γ and γ

And $\alpha\beta = 63$

$\alpha\beta\gamma^2 = 7$ so $63\gamma^2 = 7$ so either $\gamma = -\frac{1}{3}$ or $\gamma = \frac{1}{3}$

$\alpha + \beta + 2\gamma = \frac{150}{9} \Rightarrow \alpha + \beta = 16 \Rightarrow \beta = 16 - \alpha$

Solving simultaneously gives:

$\alpha(16 - \alpha) = 63$

$\alpha^2 - 16\alpha + 63 = 0$

$(\alpha - 7)(\alpha - 9) = 0$

$\alpha = 7$ or $\beta = 9$

So the solutions are 7, 9, $\frac{1}{3}$ and $\frac{1}{3}$

71) b)

$(x - 7)(x - 9)(3x - 1)(3x - 1)$

$= (x^2 - 16x + 63)(9x^2 - 6x + 1)$

$= 9x^4 - 6x^3 + x^2 - 144x^3 + 96x^2 - 16x + 567x^2 - 378x + 63$

$= 9x^4 - 150x^3 + 664x^2 - 394x + 63$

$\therefore p = 664$ and $q = 394$

72)

Roots are: α, $\beta = k\alpha$, $\gamma = k^2\alpha$ and $\delta = k^3\alpha$

$c = \alpha(k\alpha) + \alpha(k^2\alpha) + \alpha(k^3\alpha) + k\alpha(k^2\alpha) + k\alpha(k^3\alpha) + k^2\alpha(k^3\alpha)$

$= \alpha^2[k + k^2 + k^3 + k^3 + k^4 + k^5]$

$= \alpha^2\left[k^3 + \sum_{r=1}^{5} k^r\right]$

So $n = 5$